

Deviation Management – Implementation Steps

Do them in order. Steps 1–14 are client only. Steps 15–22 need a rebuild and redeploy. Steps 23–25 add the tabs.

Client – no server needed

Step 1 — WCModelDisplayProperty

Replace the constant list with this.

```
COMPONENT("MonitoredComponents", 200),
SEVERITY("Severity", 80),
LEVEL("ThresholdLevel", 60),
TYPE("Type", 80),
VALUE("ThresholdValue", 60),
MIN_VALUE("LearningThresholdMinValue", 70),
MAX_VALUE("LearningThresholdMaxValue", 70),
LEARN_TIME("healthAlarmPageColumn8", 130),
WORKCARD_REF("WorkCardRef", 130),
MIN_DEV("LearningThresholdMinDev", 140),
MEAN_MULT_FACTOR("LearningThresholdMeanMultFactor", 170),
DEV_COEF("LearningThresholdDevCoef", 130),
NB_POINTS("NbSuccessivePtForAlarmRaising", 190),
MAX_TIME_WO_ACQ("MaxTimeWithoutAcquisition", 190),
FILTERING_LAW("FilteringLaw", 110),
TE("TeReference", 140);
```

Step 2 — WCModelDisplayProperty → fromKey

Replace `return null;` at the end with this.

```
throw new IllegalArgumentException("Unknown display property key: " + property);
```

Step 3 — WCTreeLabelProvider → getColumnImage

Delete everything inside the method. Leave only this.

```
return null;
```

Step 4 — WCTreeLabelProvider → getColumnText

In the `WCThresholdModel` branch, after case `NB_POINTS`: block, before `default` :, add this.

```
case FILTERING_LAW:
    return model.getFilteringLawName() != null ? model.getFilteringLawName() : "";
case TE:
    return model.getTeReference() != null ? model.getTeReference() : "";
```

Step 5 — WCTreeLabelProvider → getBackground

Replace the whole method with this.

```
@Override
public Color getBackground(Object element, int column) {
    if (element instanceof WCThresholdModel model) {
        WCModelDisplayProperty property = WCModelDisplayProperty.values()[column];

        if (property == WCModelDisplayProperty.SEVERITY) {
            if (model.getSeverity() == Severity.RED) {
                return CoreResourcesUtils.getColor(CoreResourcesUtils.COLOR_RED);
            } else if (model.getSeverity() == Severity.AMBER) {
                return CoreResourcesUtils.getColor(CoreResourcesUtils.COLOR_AMBER);
            }
            return null;
        }
        if (model.isDeviated()) {
            return CoreResourcesUtils.getColor(CoreResourcesUtils.COLOR_GREEN);
        }
        return model.isOdd() ? ODD_NUMBER_COLOR : EVEN_NUMBER_COLOR;
    }
    return null;
}
```

Step 6 — WCTreeCellModifier → canModify

Find `if (element instanceof WCThresholdModel)`. Change that line and add the guard directly after it.

```
if (element instanceof WCThresholdModel thresholdModel) {
    if (thresholdModel.isDeviated()) {
        return false;
    }
    switch (WCModelDisplayProperty.fromKey(property)) {
```

Step 7 — WCTreeCellModifier → getValue

In the `WCThresholdModel` branch, before `default :`, add this. Also delete the commented-out `CHECKBOX` lines.

```
case FILTERING_LAW:
    return thresholdModel.getFilteringLawName() != null
        ? thresholdModel.getFilteringLawName() : "";

case TE:
    return thresholdModel.getTeReference() != null
        ? thresholdModel.getTeReference() : "";
```

Step 8 — WCTreeCellModifier → modify

Delete the whole case `CHECKBOX` block. Before `default :`, add this.

```
case FILTERING_LAW:
    if (value != null && !value.toString().trim().isEmpty()) {
        thresholdModel.setFilteringLawName(value.toString());
    } else {
        thresholdModel.setFilteringLawName(null);
    }
    break;
case TE:
    if (value != null && !value.toString().trim().isEmpty()) {
        thresholdModel.setTeReference(value.toString().trim());
    } else {
        thresholdModel.setTeReference(null);
    }
    break;
```

Step 9 — HealthAlarmsPreferenceStore

Add this method anywhere in the class.

```
public List<String> getFilteringLawNames(Aircraft aircraft) {
    List<String> names = new ArrayList<>();
    try {
        HealthMonitoringConfiguration configuration = getConfiguration(aircraft);
        for (FilteringLaw law : configuration.getFilteringLaws()) {
            names.add(law.getName());
        }
    } catch (EntityNotFoundException e) {
        LOGGER.warn("Could not load filtering laws", e);
    }
    names.sort(String::compareTo);
    return names;
}
```

Step 10 — HealthAlarmsPreferenceStore → createAllMechanicalComponents

Find the line starting `List<WCThresholdModel> thresholdModels = getThresholdModels(` and replace it with this.

```
List<WCThresholdModel> thresholdModels = getThresholdModels(healthMechanicalComponent,
    indicatorModels, entry.getValue().getIndicatorConfigurations());
```

Step 11 — HealthAlarmsPreferenceStore → getThresholdModels

Add the third parameter to the signature, and set the law inside the `for (Indicator indicator : sortedSet)` loop.

```
private List<WCThresholdModel> getThresholdModels(
    HealthMechanicalComponent healthMechanicalComponent,
    Map<Indicator, IndicatorModel> indicatorModels,
    Map<Indicator, IndicatorConfiguration> indicatorConfigurations) {

    // inside the loop, before hmcThresholdModels.addAll(...)
    IndicatorConfiguration indicatorConfiguration = indicatorConfigurations.get(indicator);
    String law = indicatorConfiguration != null ? indicatorConfiguration.getFilteringLaw() : null;
    for (WCThresholdModel thresholdModel : indicatorThresholdsMap.values()) {
        thresholdModel.setFilteringLawName(law);
        thresholdModel.setDeviated(false); // TODO BCR-6605 set true to test the green row
    }
}
```

Step 12 — HealthAlarmsPreferenceStore → isValidData

Inside for (WCThresholdModel currentModel : ...), after the switch closes, add this.

```
if (currentModel.isDeviated() && (currentModel.getTeReference() == null
    || currentModel.getTeReference().trim().isEmpty())) {
    MessageDialog.openError(parent.getShell(),
        Messages.CORE.getString("healthAlarmPageDataInvalidTitle"),
        MessageFormat.format(Messages.CORE.getString("deviationErrorTeMandatory"),
            currentModel.getIndicator().getName()));
    return false;
}
```

Step 13 — WCModelTreeView

Add the field at the top of the class, replace the FILTERING_LAW editor line in createCellEditors, and add two lines to setSelectedAircraft.

```
private ComboBoxViewerCellEditor filteringLawEditor;

// in createCellEditors, replacing:
// editors[WCModelDisplayProperty.FILTERING_LAW.ordinal()] = getCellEditor(tree, true);
filteringLawEditor = new ComboBoxViewerCellEditor(tree, SWT.READ_ONLY);
filteringLawEditor.setLabelProvider(new LabelProvider() {
    @Override
    public String getText(Object element) {
        return element == null ? "" : element.toString();
    }
});
filteringLawEditor.addListener(createWCColumnListener());
filteringLawEditor.setContentProvider(new ArrayContentProvider());
editors[WCModelDisplayProperty.FILTERING_LAW.ordinal()] = filteringLawEditor;

// in setSelectedAircraft, after this.selectedAircraft = selectedAircraft;
if (filteringLawEditor != null && selectedAircraft != null) {
    filteringLawEditor.setInput(healthAlarmsPreferenceStore.getFilteringLawNames(selectedAircraft));
}
```

Step 14 — messages.properties – and every language file

Add these keys.

```
FilteringLaw=Filtering Law
TeReference=TE *
deviationErrorTeMandatory=Error : deviation can't be set on indicator {0}. Reason : TE field is mandatory
deviationSuccess=Deviation successfully set on indicator {0}
deviationAlreadySet=A deviation is already set on this indicator.
```

Run the app now. Columns are wide, no tick box, Filtering Law shows a value and a dropdown, TE is editable. Set deviated to true on one row to check the green highlight, then set it back.

Server – needs rebuild and redeploy

Step 15 — ThresholdConfiguration (mgs-jpa-model/src/main/java)

After the workcard field, add these.

```

@Column(name = "TE_REFERENCE")
private String teReference;

@Temporal(TemporalType.TIMESTAMP)
@Column(name = "DEVIATION_START_DATE")
private Date deviationStartDate;

@Temporal(TemporalType.TIMESTAMP)
@Column(name = "DEVIATION_END_DATE")
private Date deviationEndDate;

public String getTeReference() { return teReference; }
public void setTeReference(String teReference) { this.teReference = teReference; }
public Date getDeviationStartDate() { return deviationStartDate; }
public void setDeviationStartDate(Date d) { this.deviationStartDate = d; }
public Date getDeviationEndDate() { return deviationEndDate; }
public void setDeviationEndDate(Date d) { this.deviationEndDate = d; }

public boolean isDeviationActive() {
    return deviationStartDate != null && deviationEndDate == null;
}

```

Step 16 — DDL / migration script

Add the three columns.

```

ALTER TABLE THRESHOLD_CONFIGURATION ADD TE_REFERENCE VARCHAR(255);
ALTER TABLE THRESHOLD_CONFIGURATION ADD DEVIATION_START_DATE TIMESTAMP;
ALTER TABLE THRESHOLD_CONFIGURATION ADD DEVIATION_END_DATE TIMESTAMP;

```

Step 17 — ThresholdConfigurationDto (mgs-services-interfaces)

After the workcard field, add these.

```

private String teReference;
private Date deviationStartDate;
private Date deviationEndDate;

public final String getTeReference() { return teReference; }
public final void setTeReference(String teReference) { this.teReference = teReference; }
public final Date getDeviationStartDate() { return deviationStartDate; }
public final void setDeviationStartDate(Date d) { this.deviationStartDate = d; }
public final Date getDeviationEndDate() { return deviationEndDate; }
public final void setDeviationEndDate(Date d) { this.deviationEndDate = d; }

```

Step 18 — IndicatorConfigurationDto (mgs-services-interfaces)

After the thresholdConfigurationDtos field, add this.

```

private String filteringLaw;

public final String getFilteringLaw() { return filteringLaw; }
public final void setFilteringLaw(String filteringLaw) { this.filteringLaw = filteringLaw; }

```

Step 19 — HealthDelegate → updateHealthConfiguration (mgs-delegate)

After IndicatorConfiguration indicatorConfiguration = indicatorConfigurationsMap.get(...); add the first line. After thresholdConfiguration.setWorkcard(...); add the other three.

```

indicatorConfiguration.setFilteringLaw(indicatorConfigurationDto.getFilteringLaw());

thresholdConfiguration.setTeReference(thresholdConfigurationDto.getTeReference());
thresholdConfiguration.setDeviationStartDate(thresholdConfigurationDto.getDeviationStartDate());
thresholdConfiguration.setDeviationEndDate(thresholdConfigurationDto.getDeviationEndDate());

```

Step 20 — WCThresholdModel

Add two fields with getters and setters.

```

private Date deviationStartDate;
private Date deviationEndDate;

```

Step 21 — HealthAlarmsPreferenceStore → innerSave

After `IndicatorConfigurationDto indicatorConfigurationDto = new IndicatorConfigurationDto(...);` add the first line. After `thresholdConfigurationDto.setWorkcard(...);` add the rest.

```
indicatorConfigurationDto.setFilteringLaw(wcThresholdModel.getFilteringLawName());

thresholdConfigurationDto.setTeReference(wcThresholdModel.getTeReference());
thresholdConfigurationDto.setDeviationStartDate(
    wcThresholdModel.getDeviationStartDate() != null
        ? wcThresholdModel.getDeviationStartDate()
        : (wcThresholdModel.isDeviated() ? new Date() : null));
thresholdConfigurationDto.setDeviationEndDate(wcThresholdModel.getDeviationEndDate());
```

Step 22 — HealthAlarmsPreferenceStore → getThresholdModels

Replace the temporary `setDeviated(false)` line from Step 11 with the real load, once the read path gives you the `ThresholdConfiguration`.

```
thresholdModel.setTeReference(thresholdConfiguration.getTeReference());
thresholdModel.setDeviationStartDate(thresholdConfiguration.getDeviationStartDate());
thresholdModel.setDeviationEndDate(thresholdConfiguration.getDeviationEndDate());
thresholdModel.setDeviated(thresholdConfiguration.isDeviationActive());
```

Rebuild in this order: `mgs-jpa-model`, `mgs-services-interfaces`, `mgs-delegate`, then the client. Redeploy the EAR.

Three tabs – client only

Step 23 — HealthAlarmsComposite – imports

Add these three imports.

```
import org.eclipse.swt.widgets.TabFolder;
import org.eclipse.swt.widgets.TabItem;
import com.eurocopter.mgs.core.presentation.i18n.Messages;
```

Step 24 — HealthAlarmsComposite → constructor

Find `createTreeViewer(this);` and replace that single line with this block. The A/C combo and the TE message stay above, untouched.

```
TabFolder tabFolder = new TabFolder(this, SWT.NONE);
tabFolder.setLayoutData(new GridData(SWT.FILL, SWT.FILL, true, true));

// Tab 1 - Configuration: holds the existing tree
TabItem configurationTab = new TabItem(tabFolder, SWT.NONE);
configurationTab.setText(Messages.CORE.getString("deviationTabConfiguration"));
Composite configurationComposite = new Composite(tabFolder, SWT.NONE);
GridLayout tabLayout = new GridLayout();
tabLayout.marginHeight = 0;
tabLayout.marginWidth = 0;
configurationComposite.setLayout(tabLayout);
configurationTab.setControl(configurationComposite);
createTreeViewer(configurationComposite);

// Tab 2 - Deviation list (empty for now)
TabItem deviationListTab = new TabItem(tabFolder, SWT.NONE);
deviationListTab.setText(Messages.CORE.getString("deviationTabList"));
deviationListTab.setControl(new Composite(tabFolder, SWT.NONE));

// Tab 3 - Deviation history (empty for now)
TabItem deviationHistoryTab = new TabItem(tabFolder, SWT.NONE);
deviationHistoryTab.setText(Messages.CORE.getString("deviationTabHistory"));
deviationHistoryTab.setControl(new Composite(tabFolder, SWT.NONE));
```

Step 25 — messages.properties – and every language file

Add these three keys.

```
deviationTabConfiguration=Configuration
deviationTabList=Deviation list
deviationTabHistory=Deviation history
```

createTreeView already takes a Composite parameter, so it needs no change. Tabs 2 and 3 are empty placeholders – the tables inside them are a separate ticket, as is the Manage Filtering Laws dialog.

Known problem, deferred

HealthAlarmsPreferenceStore.innerSave and getComponentThresholds both skip rows where isEnabled() is false, and WCThresholdModel defaults it to FALSE. With the tick box gone the user cannot enable a row, so a deviation on an unticked row will not save. Mark both call sites with a TODO.